

Garrett Fincke

Pittsburgh, PA | garrettfinke@gmail.com | fincke.dev | linkedin.com/in/garrett-fincke | github.com/ggfincke

Education

University of Pittsburgh

Master of Science in Computer Science

Aug 2026 – Expected May 2028

Pittsburgh, PA

The Pennsylvania State University

Bachelor of Science in Computer Science

Aug 2021 – Dec 2024

University Park, PA

Experience

Westinghouse Electric Company

Software Developer Intern, GIC Tools and Applications

May 2026 – Aug 2026

Cranberry Township, PA

- Building a Python/FastAPI service over an isolated Oracle Document Index, exposing baseline, release, and configuration-item data to engineering and AI workflows
- Implementing Active Directory basic auth, project-level Oracle authorization, API keys, rate limiting, and concurrent JSON access for restricted-data environments
- Integrating with an internal agentic AI platform (Databricks lakehouse) as a source-grounded retrieval tool for LLM-assisted I&C engineering workflows

Scale AI

Software Engineer (contract)

May 2024 – Jul 2025

Remote

- Built Python evaluation harnesses for LLM-generated code, compiling and executing submissions in sandboxed Docker environments and scoring correctness against automated tests
- Automated linting, test orchestration, and metrics pipelines to track pass@k, syntax/style errors, and failure modes across code-generation tasks
- Created schema-versioned data pipelines using Pydantic, JSON Schema, and deterministic sampling, replacing ad hoc manual review with reproducible scripted validation

Projects

SwimMate v2 | *Python, Django REST, PostgreSQL/pgvector, OpenAI, Swift, SwiftUI, iOS/watchOS*

Full-stack swimming platform with native iOS/watchOS apps and AI-generated workouts

- Built a RAG workout-generation pipeline over pgvector with JWT auth, constraint extraction, metadata reranking, role composition, provider abstraction, and generation audit logs
- Added an offline eval harness with schema-versioned corpora, deterministic validation, prompt/model/provider grouping, latency/token metrics, regression-diff replay, and a hybrid lexical+dense (RRF) retrieval path

Coral | *TypeScript, Ollama, React Ink, Commander, Node.js, Vitest, ESLint, Prettier*

Local-first CLI/TUI coding agent running entirely on Ollama models for private, on-device inference

- Built a multi-turn agent loop that cycles model reasoning and tool calls (read, write, bash, grep, glob, git) until task completion, feeding structured tool results back into conversation context
- Implemented session persistence, conversation compaction, and per-model dynamic context sizing to keep long-running tasks within each model's token budget

MDX Preview for VS Code | *TypeScript, React 19, MDX 3, VS Code Extension API, Vite 7, esbuild*

Published VS Code/Open VSX extension (7,000+ installs) for live MDX previews

- Architected a 6-package monorepo with a dual-build pipeline (esbuild extension host, Vite 7 React 19 webview) and extracted reusable runtime/compiler code into a standalone mdx-forge npm library
- Built trust-gated two-mode rendering (Safe Mode: DOMPurify-sanitized HTML under a strict CSP; Trusted Mode: compiled JS for full React 19) plus an mdx-forge-render MCP server

Technical Skills

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, C/C++

Backend & Data: Django, FastAPI, Node.js, PostgreSQL, Oracle, Redis, Celery, Pydantic, REST APIs

AI/LLM: RAG, embeddings/vector search, agentic AI, MCP, Ollama, vLLM, OpenAI/Anthropic APIs

Frontend & Mobile: React, Next.js, SwiftUI, UIKit, Tailwind CSS

Cloud & DevOps: Docker, Docker Compose, AWS EC2/SSM/CloudWatch, GitHub Actions, CI/CD, Vercel